Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
Black: Molex 2147601225
Red: Molex 2147602225

Reverse Polarity Protection

Q14
CSD19534Q5A

J10
POWER
V_IN
V_IN_RTN
Molex
0436500200

D1
SMDJ24CA
24V

C64
10uF-50V-10%-X7R-1210

C62
0.1uF-50V-10%-0603

U11
LM74700QDBVRQ1
VCAP
EN
GND

C55
10uF-50V-10%-X7R-1210

R29
5.1k-1%-0603

R31
5.1k-1%-0603

R25
5.1k-1%-0603

D2
PWR

VIN_PS

File: buck-boost-power-supply.kicad_sch
2/5
+V_IN
-V_IN
15V 3A Buck-Boost Power Supply
+V_OUT
-V_OUT

Power Supply Back
Feed Protection

Q1
CSD19534Q5A

C1
10uF-50V-10%-X7R-1210

C15
0.1uF-50V-10%-0603

U2
LM74700QDBVRQ1
VCAP
EN
GND

C16
10uF-50V-10%-X7R-1210

J4
V_OUT
V_OUT_RTN
15V
Molex
0436500200

Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
Black: Molex 2147601225
Red: Molex 2147602225

Mating Connector:
Housing: Molex 5040510401
Precrimped 24AWG wire 150mm: Molex 0797581003

J7
USB1
Molex
5040500491

File: canable2.kicad_sch
3/5
CANable2.0
USB_DP
USB_DM
3V3_USB
GND_CAN
CAN_HI
CAN_LO

J9
CAN1
MFG P/N: 0705550036
MFG: Molex

J1
Terminal

J2
Terminal
Keystone Electronics
7774

File: NX-Button-Interface.kicad_sch
4/5
NX-Button Interface

File: usb-serial.kicad_sch
5/5
USB Serial
+5V J401
3V3 J401
RXD
TXD
RESET
FORCE_RECOVERY
VDD_5V_SYS
3V3_BTN
UART2_TXD
UART2_RXD
SYS_RESET
REC
VDD_5V_SYS
NX_PC_LED-

PH1RB-15-UA
Adam Tech
15 3V3_BTN
14
13
12 NX_PC_LED-
11 VDD_5V_SYS
10 UART2_RXD
9 UART2_TXD
8 AC OK
7 Auto Power On Disable
6
5 SYS_RESET
4
3 REC
2 GND
1 SLEEP/WAKE*
NX CONTROL & UART
J3
GND1

D3
PC

SW2
RECOVERY

SW1
RESET

GND1

Light Pipe
LP1
Bivar
SLP3-150-100-F

Light Pipe
LP2
Bivar
SLP3-150-100-F

Light Pipe
LP3
Bivar
SLP3-150-100-F

Light Pipe
LP4
Bivar
SLP3-150-100-F

H3
MountingHole
H10
MountingHole
H11
MountingHole
H4
MountingHole

H5
MountingHole
H7
MountingHole
H8
MountingHole

H1
1.152mm ToolingHole
H12
1.152mm ToolingHole
H2
1.152mm ToolingHole
H13
1.152mm ToolingHole

FID1
Fiducial
FID2
Fiducial
FID3
Fiducial
FID4
Fiducial
FID5
Fiducial
FID6
Fiducial

SPACE COOKIES LOGO

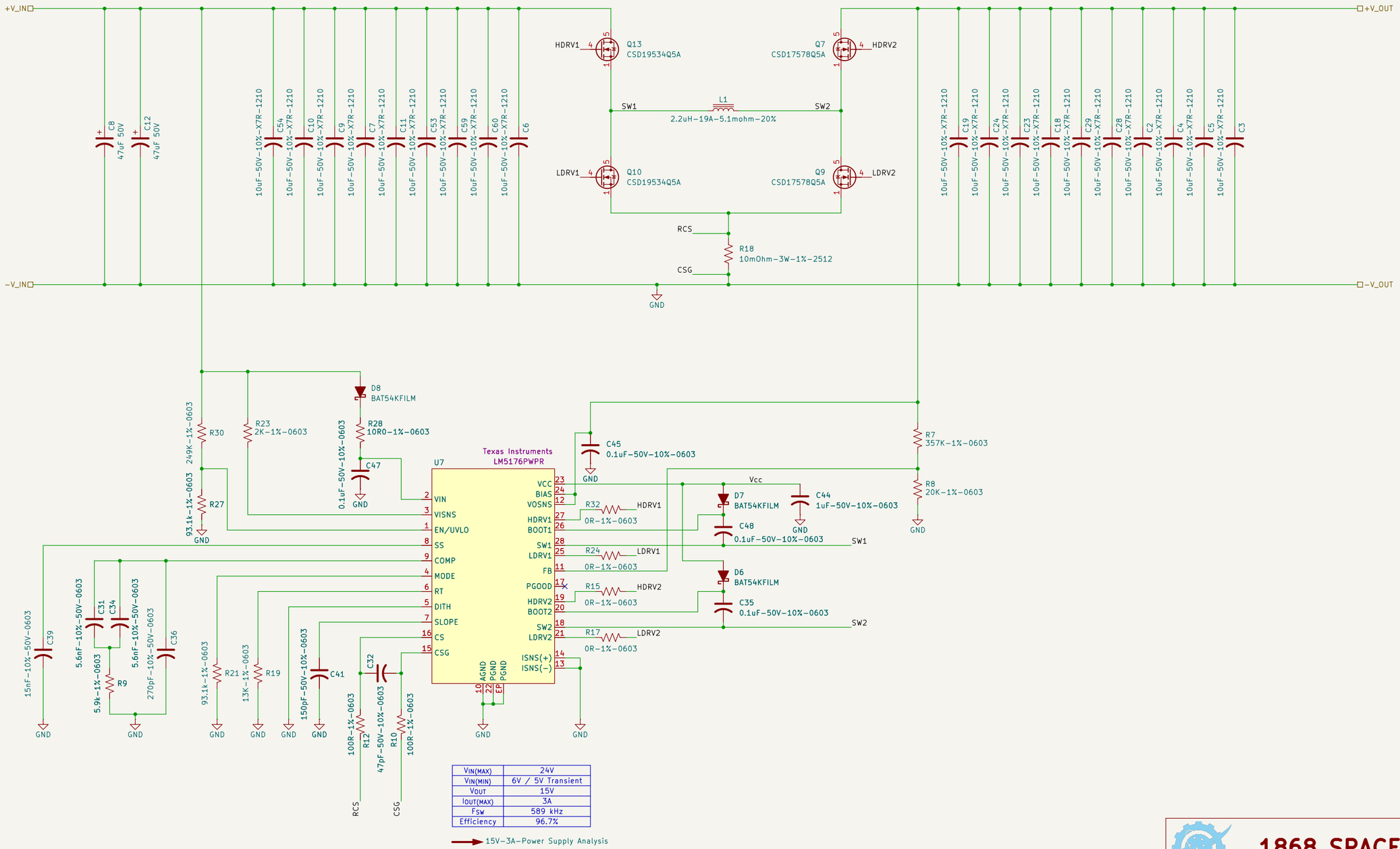
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1868 SPACE COOKIES

Sheet: /
File: TVS-Adapter-v1.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-11-01	Rev: 1
KiCad E.D.A. 10.0.8		Id: 1/5



VIN(MAX)	24V
VIN(MIN)	6V / 5V Transient
VOUT	15V
IOUT(MAX)	3A
FSW	589 kHz
Efficiency	96.7%

15V−3A−Power Supply Analysis

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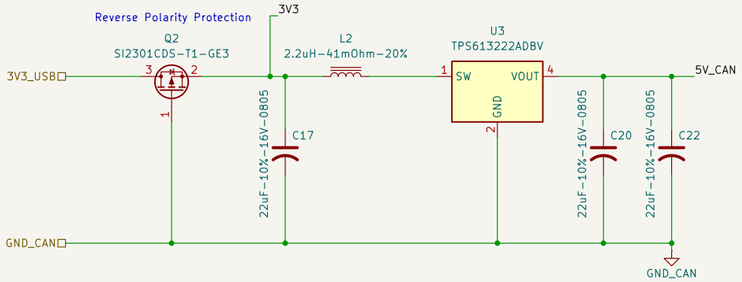
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File: buck−boost−power−supply.kicad_sch

Title: The Vision System Adapter

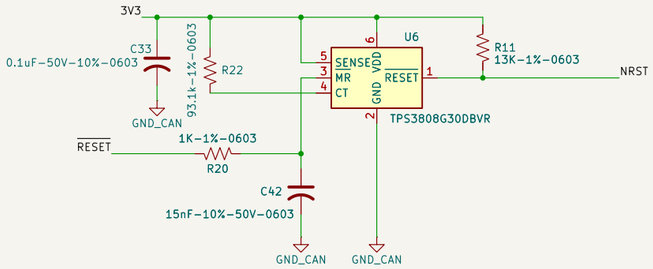
Size: B	Date: 2026−11−01	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 2/5

5V-250mA-Power Supply Analysis

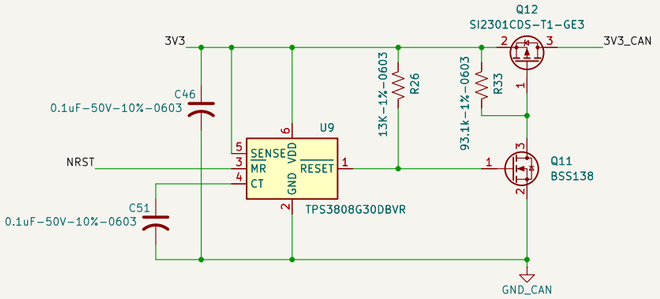
Reverse Polarity Protection

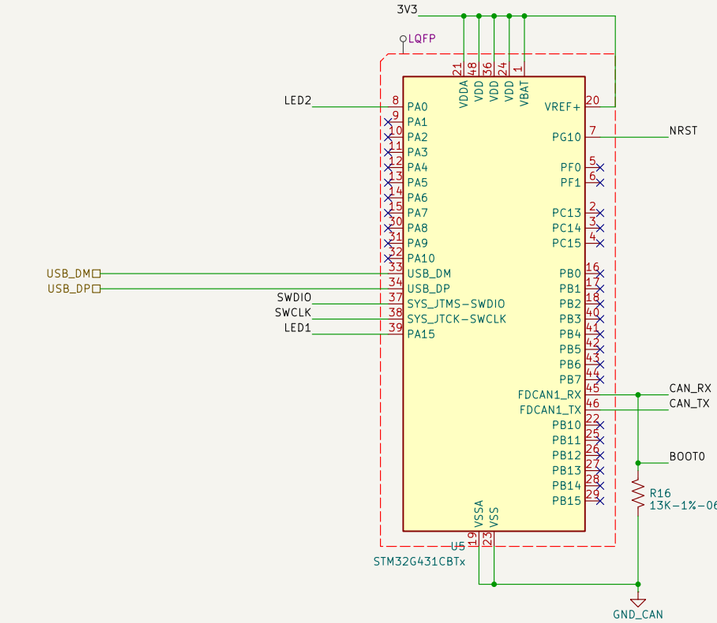


POWER ON RESET
300ms

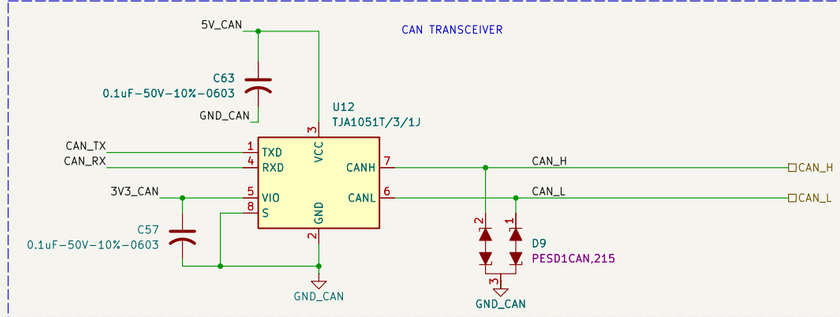


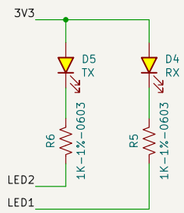
CAN BOOTO POWER DELAY
0.5 seconds

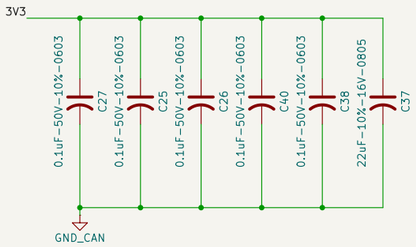




CAN TRANSCEIVER







DEBUG HEADER



NOTES:
- CANABLE circuit interfaces with the M.2 Key-E adapter plugged into the J401. Input power must be no greater than 3.6V

CANable 2.0 Schematic

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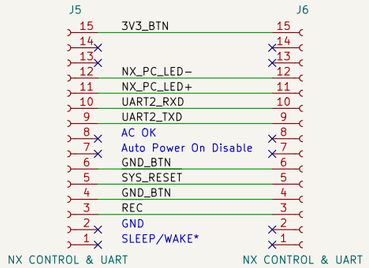
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Sheet: /CANable2.0/
File: canable2.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 3/5



H6
M3.7 MountingHole

H9
M3.7 MountingHole

SPACE COOKIES LOGO

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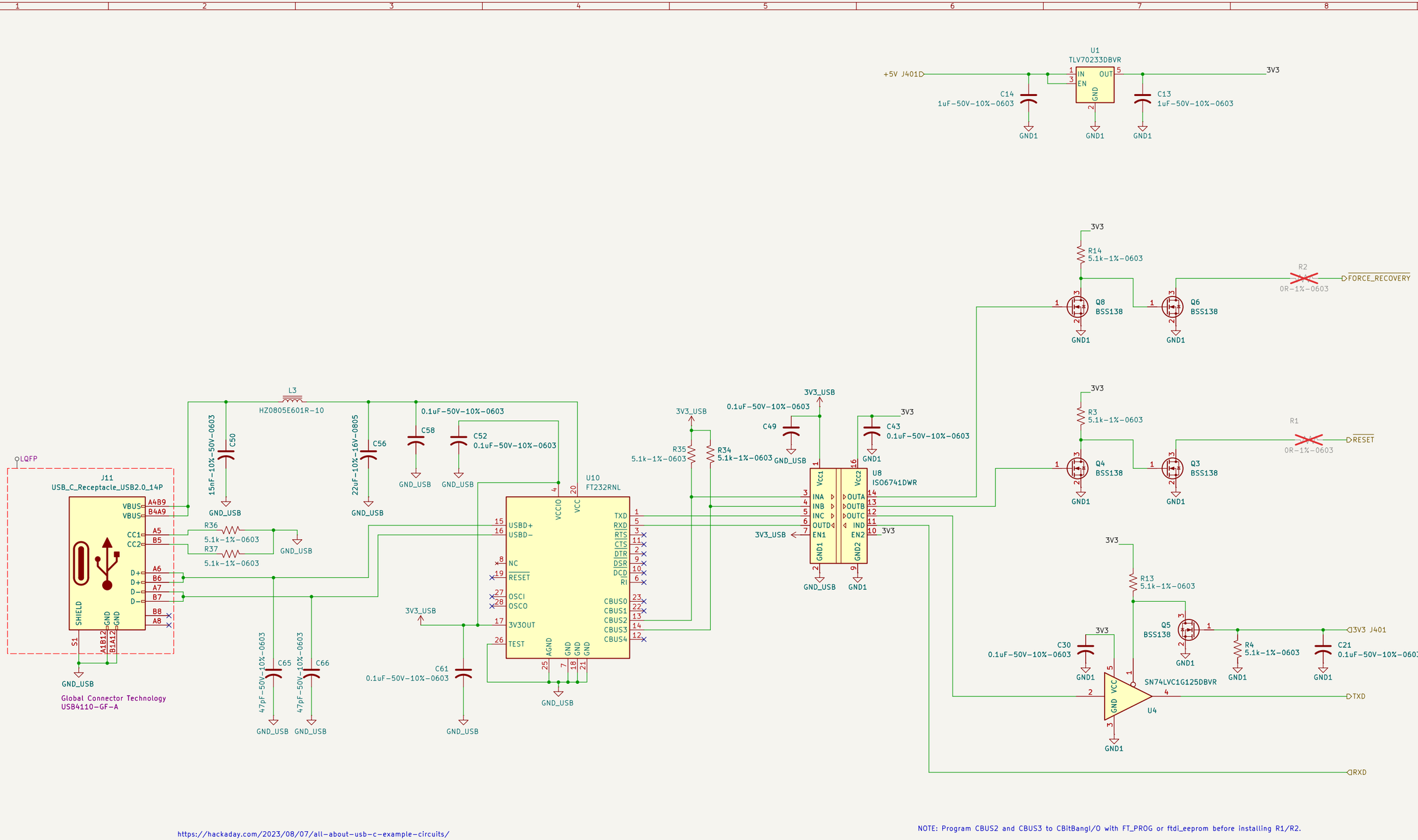
open source hardware

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Sheet: /NX-Button Interface/
File: NX-Button-Interface.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 4/5



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Sheet: /USB Serial/
File: usb-serial.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 5/5